

GenAI Disclosure

What type of GenAI did you use for this assignment?

Gemini and ChatGPT.

What did you use GenAI for in this assignment?

- **Brainstorming and Idea Generation:** Collaborated with GenAI to refine the project scope from broad interests (demographics, silver economy, and social media) into a specific, novel research problem regarding infrastructure mismatch. Discussed data accessibility and feasibility.
- **Literature Exploration Support:** Used GenAI to identify and recommend relevant academic papers and reports (e.g., UNICEF, BMC Geriatrics) to validate the proposal and establish the "Related Work" section.
- **Writing, Structuring, and Clarity Improvement:** Utilized GenAI for proofreading, optimizing academic expression, and trimming redundant content to adhere to the 1,000-word limit.

Demographic Structure and Public Service Resource Distribution Across Chinese Provinces

Introduction

China is undergoing a rapid demographic transition characterised by declining fertility and accelerating population ageing. According to UNICEF (2024), births dropped from 18.83 million in 2016 to 9.02 million in 2023, while the proportion of the population aged 60 and above reached 21.1 percent. These shifts may significantly influence the demand for public services, particularly education and healthcare, which are essential components of social welfare and human development.

At the same time, regional economic development in China remains uneven. Disparities in income and infrastructure across provinces can lead to unequal access to public services. As demographic structures change, such inequalities may intensify, creating mismatches between service capacity and population needs.

This project investigates whether demographic structure is associated with differences in the availability of healthcare and education resources across China's 31 provinces. Using provincial-level data from the National Bureau of Statistics between 2016 and 2024, the study examines relationships between key demographic indicators, including fertility rates and ageing ratios, and selected measures of public service capacity. The goal is to explore spatial disparities and identify potential patterns in resource distribution, providing an empirical perspective on how demographic transition may influence regional public service allocation.

Related Work

Existing studies have examined spatial inequalities in the distribution of public service resources in China. Using geographic information systems (GIS) and spatial statistical methods. Ma et al. (2017) reported that hospital resources are unevenly distributed across provinces, with a concentration of high-quality facilities in economically developed eastern regions. Similarly, research suggests that residents in less developed regions often face higher travel time and accessibility barriers when seeking health and education services.

Zhang et al. (2017) quantified healthcare inequality using Gini coefficients, revealing significant disparities in the geographic distribution of healthcare institutions, workers, and hospital beds. Their findings indicate that wealthier populations tend to access better-resourced hospitals, while poorer populations rely more heavily on under-resourced primary care institutions. More recent work by Song et al. (2024) confirms the persistence of spatial disparities at the county level, identifying regional hotspots of healthcare resource concentration that have changed little over time.

At the same time, Demographic change has received growing scholarly attention.. Niu and Qi (2020) found a strong negative association between women's education and fertility across different cohorts and provinces, highlighting structural changes in population dynamics. Yang (2025) also notes that ageing populations and low fertility rates are creating increasing pressure on labour markets, pension systems, and healthcare financing.

Demographic changes may also influence the distribution of education resources. Guo (2025) examined the relationship between urbanisation, changes in the school-age population, and the allocation of compulsory education resources in China. The study found that although education funding and teacher numbers have

increased overall, significant disparities remain between urban and rural regions. These findings suggest that both demographic and socioeconomic factors influence regional education resource distribution.

Despite this body of work, relatively few studies have examined how demographic structure simultaneously relates to both healthcare and education resource availability across provinces. Existing studies tend to treat these domains in isolation and rarely integrate demographic indicators with multi-sector public service data within a unified analytical framework. This project addresses that gap by combining demographic and public service data to examine cross-sectoral resource mismatches — an approach not adopted in prior provincial-level studies.

Business Model

Public service planning in China faces increasing pressure due to rapid demographic change and regional development disparities. Policymakers must allocate limited healthcare and education resources in response to declining fertility rates, an ageing population, and uneven economic development. However, decisions regarding infrastructure investment and service distribution are often made with limited understanding of how demographic structure relates to regional resource availability.

From a data-driven perspective, this project can be viewed as an information provider model, where data analysis is used to generate insights that support evidence-based decision-making.

The data scientist's role is to integrate provincial demographic statistics with public service metrics from the NBS, apply statistical and spatial analysis techniques, and translate findings into interpretable insights for non-technical stakeholders. Outputs will provide analytical results that help stakeholders better understand spatial inequalities and demographic pressures across provinces.

The analytical pipeline will involve both descriptive and exploratory analytics. Descriptive analysis will identify patterns in demographic structure and public service resource distribution, while exploratory statistical analysis - such as correlation analysis and spatial regression - may reveal relationships between demographic characteristics and service capacity. These insights may help policymakers identify provinces where healthcare and education resources are not aligned with demographic structure, such as regions with rapidly ageing populations but insufficient healthcare capacity, or regions with declining birth rates but relatively high education resource allocation. This enables more targeted and evidence-based decisions in resource planning and infrastructure investment.

The primary stakeholders include national policymakers, provincial planning agencies, public health authorities, and education administrators responsible for infrastructure investment and service allocation. The findings may also inform businesses in related sectors by highlighting emerging demand gaps. Overall, the insights generated by this project may support more informed decisions regarding regional investment priorities and long-term strategies for managing demographic transition.

Reference

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